$\Delta G^\circ = -nFE^\circ$

$= -2 \times 96500 \times 3.17$

$= -611810$ kJ

$\Delta G^o = -6.12 \times 10^5$ J

$W = 6.12 \times 10^5$ J

$\Delta G^o = -2.303\ RT \log^K C$

$\Rightarrow \log K_c = \dfrac{6.12 \times 10^5}{2.303 \times 8.314 \times 298}$

K_c = Antilog of (107.2)

23. Electrolysis of water

At anode:

$2H_2O \rightarrow 4H^+ + O_2 + 4e^-$.....(1)

At cathode:

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

Overall reaction $6H_2O \rightarrow 4H^+ + 4OH^- + 2H_2 + O_2$

(or)

Equation (1) +(2) $\times 2 \Rightarrow 2H_2O \rightarrow 2H_2 + O_2$

$\therefore$ According to faradays Law of electrolysis, to electrolyse two mole of Water $(36g \simeq 36\ mL\ of\ H_2O)$, 4F charge is required alternatively, when 36 mL of water is electrolysed, the charge generated = 4×96500 C.

$\therefore$ When the whole water which is available on the lake is completely electrolysed the amount of charge generated is equal to $\dfrac{4 \times 96500\ C}{36\ mL} \times 9 \times 10^{12} L$

$= \dfrac{4 \times 96500 \times 9 \times 10^{12}}{36 \times 10^{-3}} C$

$= 96500 \times 10^{15}$ C

$\therefore$ Given that in 1 second, 2×10^6 C is generated therefore, the time required to generate

96500×10^{15} C is $= \dfrac{1\ S}{2 \times 10^6 C} \times 96500 \times 10^{15} C$

$= 48250 \times 10^9$ S

$\therefore$ Number of years $= \dfrac{48250 \times 10^9}{365 \times 24 \times 60 \times 60}$

$= 1.5299 \times 10^6$ years

1 year = 365 days

$= 365 \times 24$ hours

$= 365 \times 24 \times 60$ min

$= 365 \times 24 \times 60 \times 60$ sec.

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